

1A: Caesar Cipher Implementation

Java

```
import java.util.*;
import java.lang.*;
import java.io.*;

class Codechef {
    private static String encrypt(String s, int key) {
        int l = s.length();
        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < l; i++) {
            if (Character.isUpperCase(s.charAt(i))) {
                char shifted = (char) (((s.charAt(i) - 'A') + key)
% 26) + 'A');
                sb.append(shifted);
            } else if (Character.isLowerCase(s.charAt(i))) {
                char shifted = (char) (((s.charAt(i) - 'a') + key)
% 26) + 'a');
                sb.append(shifted);
            } else {
                sb.append(s.charAt(i));
            }
        }
        return sb.toString();
    }

    private static String decrypt(String s, int key) {
        return (encrypt(s, 26 - (key % 26)));
    }

    public static void main(String[] args) throws
java.lang.Exception {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
    }
}
```

```

        int key = sc.nextInt();
        s = encrypt(s, key);
        System.out.println("Encrypted: " + s);
        System.out.println("Decrypted: " + decrypt(s, key));
    }
}

```

1B: Monoalphabetic Substitution Cipher

Java

```

import java.util.*;
import java.lang.*;
import java.io.*;

class Codechef {
    private static String encrypt(String s) {
        String a = "abcdefghijklmnopqrstuvwxyz";
        String b = "zyxwvutsrqponmlkjihgfedcba";

        int l = s.length();
        String encrypt = "";
        for (int i = 0; i < l; i++) {
            int j = a.indexOf(s.charAt(i));
            encrypt += (b.charAt(j));
        }
        return encrypt;
    }

    public static void main(String[] args) throws
java.lang.Exception {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        s = encrypt(s);
    }
}

```

```

        System.out.println("Encrypted: " + s);
        System.out.println("Decrypted: " + encrypt(s));
    }
}

```

1C.

2: Implement symmetric block cipher encryption and decryption using DES algorithm in C/JAVA

Java

```

import java.util.*;
import java.security.*;
import java.lang.*;
import java.io.*;
import javax.crypto.*;

class Codechef {
    private static Cipher cipher;
    static Scanner sc = new Scanner(System.in);

    private static byte[] encrypt(String data) throws Exception {
        byte[] encrypt = cipher.doFinal(data.getBytes());
        System.out.println("Encrypted: " +
Arrays.toString(encrypt));
        return encrypt;
    }

    private static byte[] decrypt(byte[] data) throws Exception {
        byte[] decrypt = cipher.doFinal(data);
    }
}

```

```

        System.out.println("Decrypted: " +
Arrays.toString(decrypt));
        return decrypt;
    }

    public static void main(String[] args) throws
java.lang.Exception {
        try {
            String data = sc.nextLine();
            KeyGenerator keygen = KeyGenerator.getInstance("DES");
            SecretKey key = keygen.generateKey();

            cipher = Cipher.getInstance("DES/ECB/PKCS5Padding");

            cipher.init(Cipher.ENCRYPT_MODE, key);
            byte[] encrypted = encrypt(data);

            cipher.init(Cipher.DECRYPT_MODE, key);
            byte[] decrypted = decrypt(encrypted);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

3: Write a C/JAVA program to implement encryption technique using Blowfish algorithm

Same code structure as Program 2, with algorithm updated to **Blowfish** (e.g., `KeyGenerator.getInstance("Blowfish")` and `Cipher.getInstance("Blowfish")`).

4: Implement the encryption of block chunk of 128 bits size using AES algorithm in C/JAVA

Same code structure as Program 2, with algorithm updated to **AES** (e.g., `KeyGenerator.getInstance("AES")` and `Cipher.getInstance("AES")`).

5: Write a C/JAVA program on Rivest Cipher 4(RC4) logic.

Java

```
import java.util.*;
import java.lang.*;
import java.io.*;

class Codechef {
    private static String encrypt(String data, String key) {
        int[] s = new int[256];
        int[] k = new int[256];
        StringBuilder encrypt = new StringBuilder();
        char[] datachars = data.toCharArray();
        char[] keychars = key.toCharArray();

        int[] dataint = new int[data.length()];
        int[] keyint = new int[key.length()];

        for (int m = 0; m < data.length(); m++) {
            dataint[m] = (int) datachars[m];
        }
        for (int m = 0; m < key.length(); m++) {
            keyint[m] = (int) keychars[m];
        }

        int j = 0;
        for (int m = 0; m < 256; m++) {
            s[m] = m;
            k[m] = keyint[m % key.length()];
        }
    }
}
```

```

        for (int m = 0; m < 256; m++) {
            j = (j + s[m] + k[m]) % 256;
            int temp = s[j];
            s[j] = s[m];
            s[m] = temp;
        }

        int i = 0;
        j = 0;

        for (int m = 0; m < data.length(); m++) {
            i = (i + 1) % 256;
            j = (j + s[i]) % 256;
            int temp = s[i];
            s[i] = s[j];
            s[j] = temp;
            int z = s[(s[i] + s[j]) % 256];
            encrypt.append(z ^ dataint[m]);
        }
        return encrypt.toString();
    }

    private static String decrypt(String data, String key) {
        return encrypt(data, key);
    }

    public static void main(String[] args) throws
    java.lang.Exception {
        Scanner sc = new Scanner(System.in);
        String data = sc.nextLine();
        String key = sc.nextLine();

        System.out.println("Encrypted: " + encrypt(data, key));
        System.out.println("Decrypted: " + decrypt(data, key));
    }
}

```

6.Implement DES-2 and DES-3 using Java cryptography package

Java

```
import java.util.*;
import java.lang.*;
import java.io.*;
import java.security.*;
import javax.crypto.*;

class Codechef {
    private static Cipher cipher;
    private static byte[] encrypt(String data, SecretKey key1,
SecretKey key2, int flag) throws Exception {
        String algo = (flag == 2) ? "DESede" : "DES";
        cipher = Cipher.getInstance(algo + "/ECB/PKCS5Padding");
        cipher.init(Cipher.ENCRYPT_MODE, key1);
        byte[] encrypted = cipher.doFinal(data.getBytes());
        if (flag == 1) {
            cipher.init(Cipher.ENCRYPT_MODE, key2);
            encrypted = cipher.doFinal(encrypted);
        }
        System.out.println("Encrypted: " + new String(encrypted));
        return encrypted;
    }

    private static byte[] decrypt(byte[] data, SecretKey key1,
SecretKey key2, int flag) throws Exception {
        String algo = (flag == 2) ? "DESede" : "DES";
        byte[] decrypted;
        cipher = Cipher.getInstance(algo + "/ECB/PKCS5Padding");
        if (flag == 1) {
            cipher.init(Cipher.DECRYPT_MODE, key2);
```

```

        decrypted = cipher.doFinal(data);
    } else {
        decrypted = data;
    }
    cipher.init(Cipher.DECRYPT_MODE, key1);
    decrypted = cipher.doFinal(decrypted);
    System.out.println("Decrypted: " + new String(decrypted));
    return decrypted;
}

public static void main(String[] args) throws
java.lang.Exception {
    try {
        Scanner sc = new Scanner(System.in);
        String data = sc.nextLine();
        int flag = sc.nextInt();
        String algo = (flag == 2) ? "DESede" : "DES";
        KeyGenerator keygen = KeyGenerator.getInstance(algo);
        SecretKey key1 = keygen.generateKey();
        SecretKey key2 = keygen.generateKey();
        byte[] encrypted = encrypt(data, key1, key2, flag);
        decrypt(encrypted, key1, key2, flag);
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

7. Write a JAVA Program to implement the RSA algorithm.

Java

```
import java.util.*;
import java.lang.*;
import java.io.*;
import java.math.*;

class Codechef {
    static Random r = new Random();
    private static BigInteger p, q;
    private static BigInteger n, phi;
    private static BigInteger e, d;
    private static int BitLength = 1024;

    private static void setter() {
        p = BigInteger.probablePrime(BitLength, r);
        q = BigInteger.probablePrime(BitLength, r);
        e = BigInteger.probablePrime(BitLength / 2, r);

        n = p.multiply(q);
        phi =
        (p.subtract(BigInteger.ONE)).multiply(q.subtract(BigInteger.ONE));

        while (phi.gcd(e).compareTo(BigInteger.ONE) > 0 &&
e.compareTo(phi) < 0) {
            e = e.add(BigInteger.ONE);
        }
    }
}
```

```

        d = e.modInverse(phi);
    }

    private static byte[] encrypt(String data) {
        BigInteger msg = new BigInteger(1, data.getBytes());
        BigInteger encrypted = msg.modPow(e, n);
        return encrypted.toByteArray();
    }

    private static byte[] decrypt(byte[] data) {
        BigInteger msg = new BigInteger(1, data);
        BigInteger decrypted = msg.modPow(d, n);
        return decrypted.toByteArray();
    }

    public static void main(String[] args) throws
java.lang.Exception {
        Scanner sc = new Scanner(System.in);
        String data = sc.nextLine();
        setter();
        byte[] encrypted = encrypt(data);
        System.out.println("Encrypted: " + new String(encrypted));
        byte[] decrypted = decrypt(encrypted);
        System.out.println("Decrypted: " + new String(decrypted));
    }
}

```

8. Implement key exchange protocol using the Diffie-Hellman algorithm.

Java

```
import java.util.*;
import java.lang.*;
import java.math.*;
import java.io.*;

class Codechef {
    public static void main(String[] args) throws
    java.lang.Exception {
        BufferedReader br = new BufferedReader(new
        InputStreamReader(System.in));

        System.out.println("Enter prime number: ");
        BigInteger p = new BigInteger(br.readLine());
        System.out.println("Enter primitive root: ");
        BigInteger g = new BigInteger(br.readLine());
        System.out.println("Enter a value x less than p: ");
        BigInteger x = new BigInteger(br.readLine());
        BigInteger r1 = g.modPow(x, p);
        System.out.println("Value of R1: " + r1);
        System.out.println("Enter a value y less than p: ");
        BigInteger y = new BigInteger(br.readLine());
        BigInteger r2 = g.modPow(y, p);
        System.out.println("Value of R2: " + r2);
        BigInteger k1 = r2.modPow(x, p);
        System.out.println("Value of K1: " + k1);
        BigInteger k2 = r1.modPow(y, p);
        System.out.println("Value of K2: " + k2);
        System.out.println("Deffie Hellman secret key has been
        taken");
    }
}
```

9. 9. Write a JAVA program to compute message digest of text using SHA-1 algorithm.

Java

```
import java.util.*;
import java.security.*;
import java.lang.*;
import java.io.*;

class Codechef {
    public static String bytesToHex(byte[] data) {
        char[] hex = {'0', '1', '2', '3', '4', '5', '6', '7', '8',
'9', 'A', 'B', 'C', 'D', 'E', 'F'};
        StringBuilder sb = new StringBuilder();
        for (int i = 0; i < data.length; i++) {
            sb.append(hex[(data[i] >> 4) & 0x0f]);
            sb.append(hex[data[i] & 0x0f]);
        }
        return sb.toString();
    }

    public static void main(String[] args) throws
java.lang.Exception {
        // your code goes here
        try {
            MessageDigest md = MessageDigest.getInstance("SHA-1");
            System.out.println("Message Digest details:");
            System.out.println("Algorithm: " + md.getAlgorithm());
            System.out.println("ToString: " + md.toString());

            String input = "";
            md.update(input.getBytes());
            byte[] output = md.digest();
            System.out.println("SHA-1(\"" + input + "\")=" +
bytesToHex(output));
        }
    }
}
```

```

        input = "abc";
        md.update(input.getBytes());
        output = md.digest();
        System.out.println("SHA-1(\"" + input + "\")=" +
bytesToHex(output));
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

10. Calculate the message digest of a text using the MD5 algorithm in JAVA.

Java

```

import java.util.*;
import java.lang.*;
import java.io.*;
import java.security.*;

class Codechef {
    public static void main(String[] args) throws
java.lang.Exception {
        // your code goes here
        try {
            Scanner sc = new Scanner(System.in);
            String s = sc.nextLine();
            MessageDigest md = MessageDigest.getInstance("MD5");
            System.out.println("Message Digest details:");
            System.out.println("Algoriithm: " + md.getAlgorithm());
            System.out.println("ToString: " + md.toString());
            md.update(s.getBytes());
            System.out.println("Input: " + s);

```

```
        System.out.println("Output: " +  
HexFormat.of().formatHex(md.digest()));  
    } catch (Exception e) {  
        e.printStackTrace();  
    }  
}  
}
```

11.Explore the java classes related to Digital Certificates.

Java

```
import java.util.*;
import java.lang.*;
import java.security.*;
import java.io.*;

class Codechef {
    public static void main(String[] args) throws
    java.lang.Exception {
        // your code goes here
        try {
            KeyPairGenerator keygen =
KeyPairGenerator.getInstance("DSA");
            keygen.initialize(1024);
            KeyPair keyPair = keygen.generateKeyPair();
            PrivateKey privateKey = keyPair.getPrivate();
            PublicKey publicKey = keyPair.getPublic();

            Scanner sc = new Scanner(System.in);
            String data = sc.nextLine();
            byte[] databytes = data.getBytes();

            Signature signature =
Signature.getInstance("SHA256withDSA");
            signature.initSign(privateKey);
            signature.update(databytes);
            byte[] digitalSignature = signature.sign();

            System.out.println("Digital Certificate: " +
Base64.getEncoder().encodeToString(digitalSignature));

            signature.initVerify(publicKey);
            signature.update(databytes);
            boolean isVerified =
signature.verify(digitalSignature);
            System.out.println("Is verified: " + isVerified);
        } catch (Exception e) {
```

```
        e.printStackTrace();
    }
}
}
```

12. Write a JAVA program to demonstrate Cross-Site Scripting (XSS) prevention.

Java

```
import java.util.*;
import java.lang.*;
import java.io.*;
import org.apache.commons.text.StringEscapeUtils;

class Codechef {
    public static void main(String[] args) throws
    java.lang.Exception {
        // your code goes here
        Scanner sc = new Scanner(System.in);
        String data = sc.nextLine();
        System.out.println("User data with xss protection: " +
data);
        String safeData = StringEscapeUtils.escapeHtml4(data);
        System.out.println("User data with xss protection: " +
safeData);
    }
}
```